



ASSIGNMENT-6

NEO-HOOKEAN MATERIAL DIRECT AND INCREMENTAL STRESS UPDATE

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Program	:	MS
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Course Name	:	Non Linear Finite Element Analysis of Solid Continua

MAP-1 (PURE TRANSLATION)

Time t = 0.78

F =
[[1. 0. 0.]
[0. 1. 0.]
[0. 0. 1.]]

C =
[[1. 0. 0.]
[0. 1. 0.]
[0. 0. 1.]]

E =
[[0. 0. 0.]
[0. 0. 0.]
[0. 0. 0.]]

b =
[[1. 0. 0.]
[0. 1. 0.]
[0. 0. 1.]]

e =
[[0. 0. 0.]
[0. 0. 0.]
[0. 0. 0.]]

S =
[[0. 0. 0.]
[0. 0. 0.]
[0. 0. 0.]]

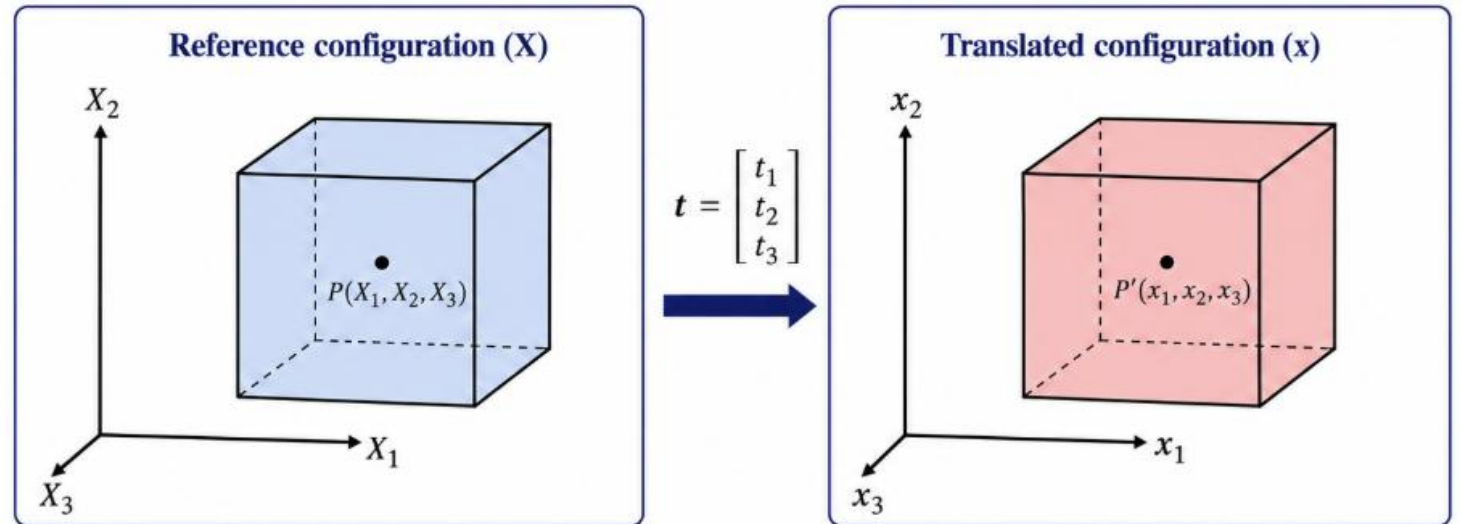
sigma =
[[0. 0. 0.]
[0. 0. 0.]
[0. 0. 0.]]

Seq = 0.0
sigma_eq = 0.0
||E|| = 0.0
||e|| = 0.0

CODE OUTPUT

Pure Translation Mapping

In pure translation, every point in the body is displaced by the same translation vector $\mathbf{t}$.
The shape and orientation remain unchanged.



Translation mapping: $\mathbf{x} = \mathbf{X} + \mathbf{t}$

In general: $x_i = X_i + t, \quad i = 1, 2, 3$

Component form:
$$\begin{cases} x_1 = X_1 + t \\ x_2 = X_2 + t \\ x_3 = X_3 + t \end{cases}$$

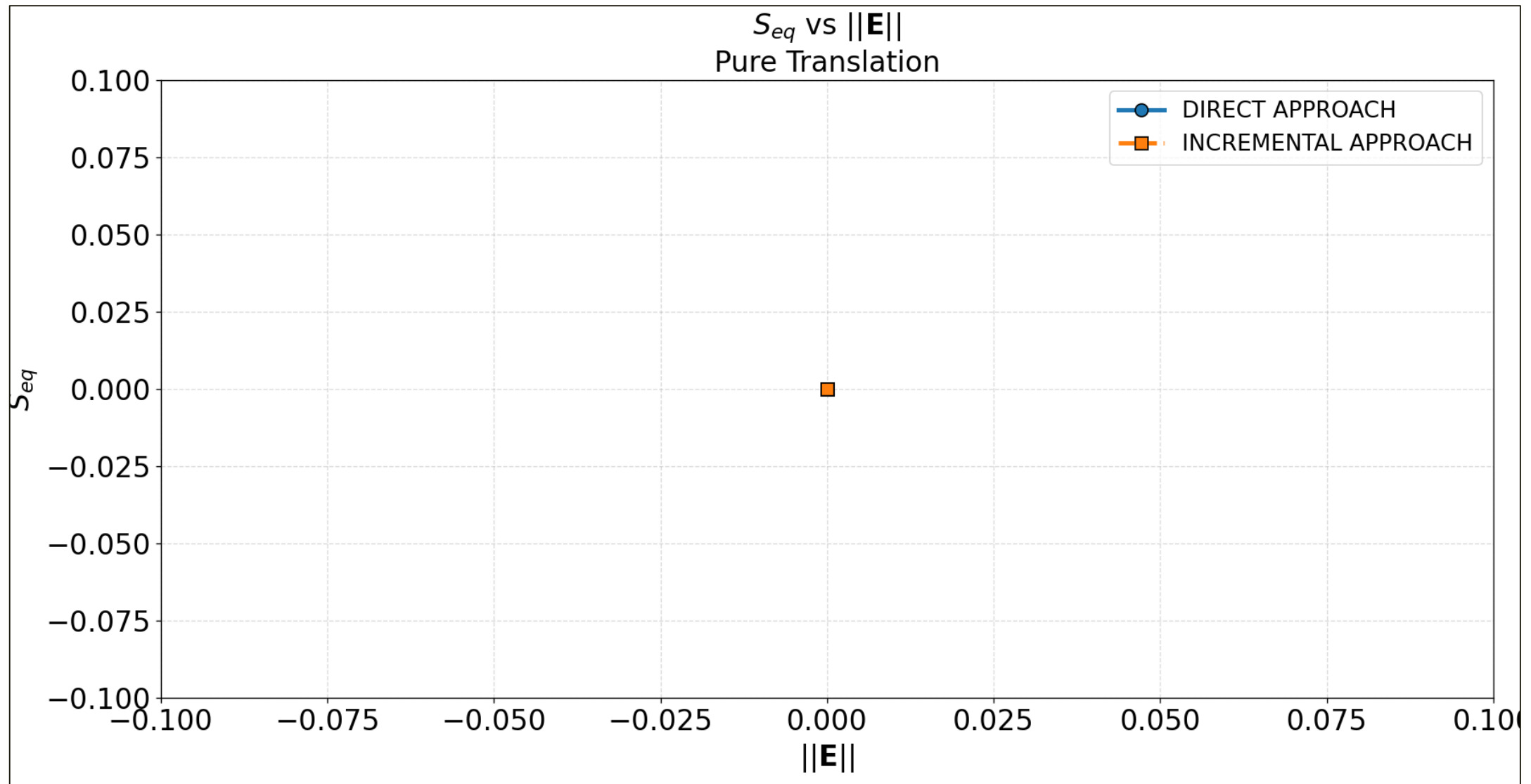
Rigid-body result:

No strain:

$\mathbf{F} = \mathbf{I}, \quad \mathbf{C} = \mathbf{I}, \quad \mathbf{E} = \mathbf{0}$

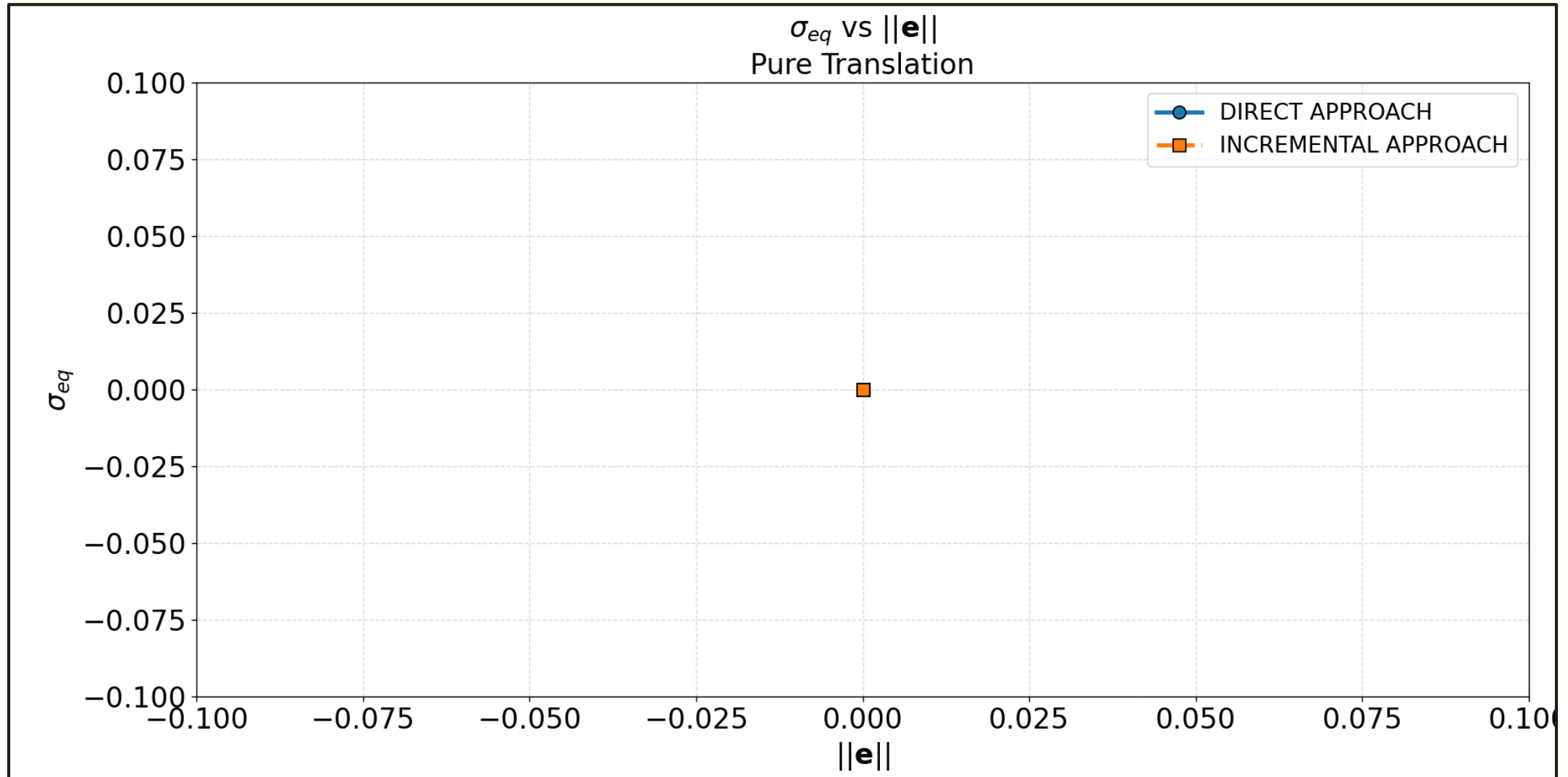


Seq vs IIEI plot





σ_{eq} vs $\|e\|$ plot





MAP-2 (TRANSLATION+ROTATION)

```
Time t = 0.99
=====
F =
[[ 0.86863151 -0.49545867  0.      ]
 [ 0.49545867  0.86863151  0.      ]
 [ 0.          0.          1.      ]]

C =
[[1.00000000e+00 8.19170499e-17 0.00000000e+00]
 [8.19170499e-17 1.00000000e+00 0.00000000e+00]
 [0.00000000e+00 0.00000000e+00 1.00000000e+00]]

E =
[[0.00000000e+00 4.09585250e-17 0.00000000e+00]
 [4.09585250e-17 1.11022302e-16 0.00000000e+00]
 [0.00000000e+00 0.00000000e+00 0.00000000e+00]]

b =
[[ 1.00000000e+00 -8.19170499e-17 0.00000000e+00]
 [-8.19170499e-17 1.00000000e+00 0.00000000e+00]
 [ 0.00000000e+00 0.00000000e+00 1.00000000e+00]]

e =
[[ 0.00000000e+00 -4.09585250e-17 0.00000000e+00]
 [-4.09585250e-17 1.11022302e-16 0.00000000e+00]
 [ 0.00000000e+00 0.00000000e+00 0.00000000e+00]]

S =
[[0.00000000e+00 7.16774187e-12 0.00000000e+00]
 [7.16774187e-12 1.94289029e-11 0.00000000e+00]
 [0.00000000e+00 0.00000000e+00 0.00000000e+00]]

sigma =
[[-1.40018333e-12 -4.71295918e-12 0.00000000e+00]
 [-4.71295918e-12 2.08290863e-11 0.00000000e+00]
 [ 0.00000000e+00 0.00000000e+00 0.00000000e+00]]

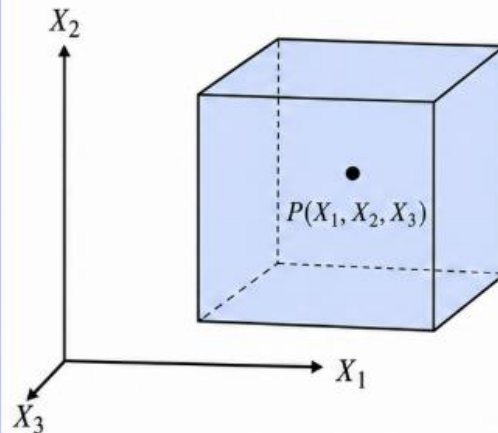
Seq = 0.0
sigma_eq = 0.0
||E|| = 0.0
||e|| = 0.0
```

CODE OUTPUT

Rotation + Translation Mapping

Rigid-body motion: rotation about the X_3 -axis combined with a uniform translation in the X_1 - X_2 plane. Shape and size remain unchanged.

Reference configuration (X)



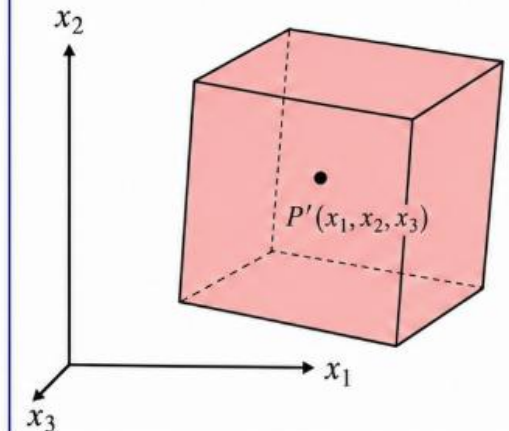
$$\theta = \left(\frac{\pi}{6}\right)t$$



translation vector
 $c(t) = (t, t, 0)^T$

$$x = R(\theta)X + c(t)$$

Current configuration (x)



Rotation about X_3 by θ :

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Component relations:

$$\begin{cases} x_1 = \cos \theta X_1 - \sin \theta X_2 + t \\ x_2 = \sin \theta X_1 + \cos \theta X_2 + t \\ x_3 = X_3 \end{cases}$$

Rigid-body result

$$F = R(\theta)$$

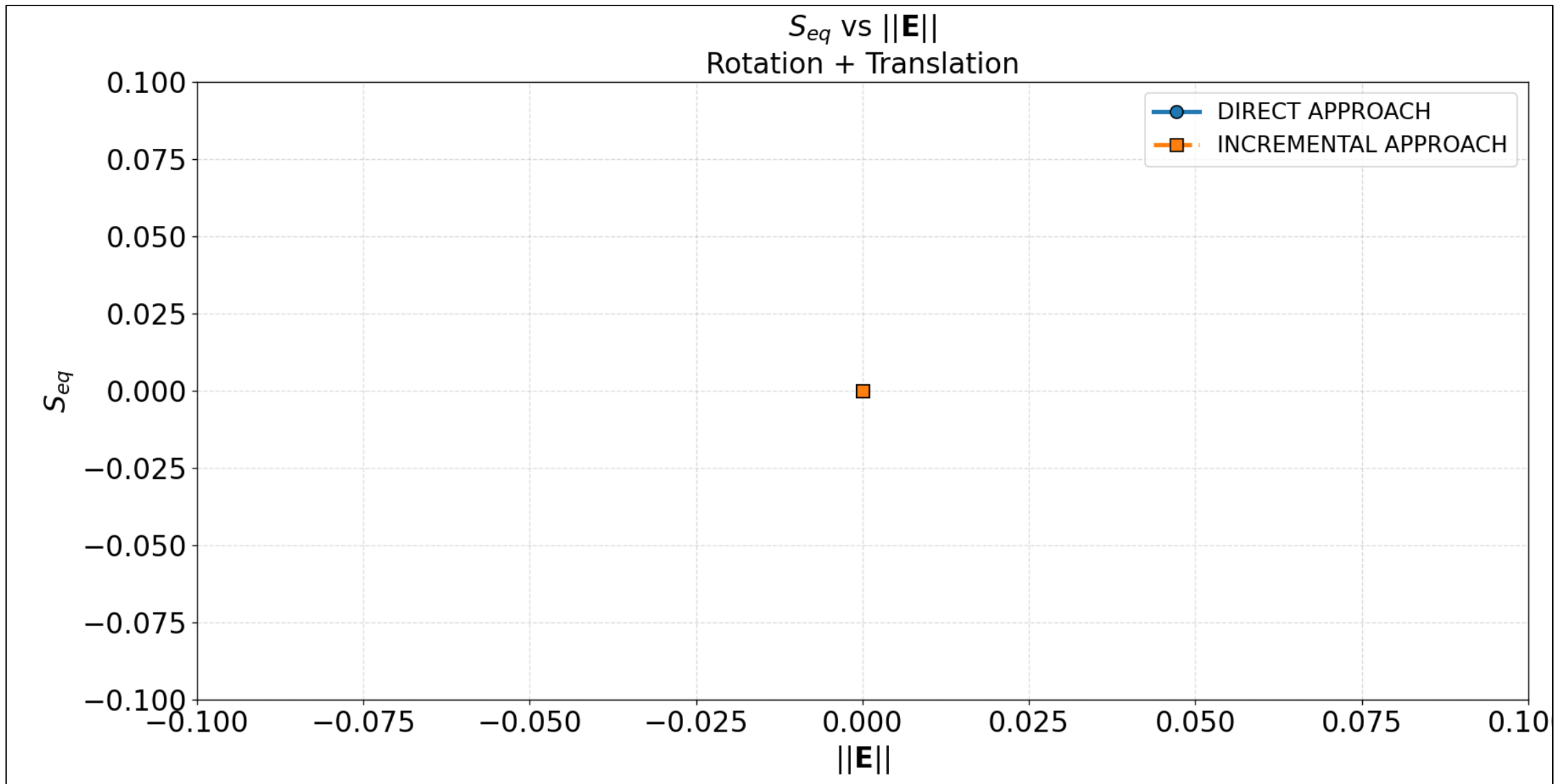
$$C = I$$

$$E = 0$$

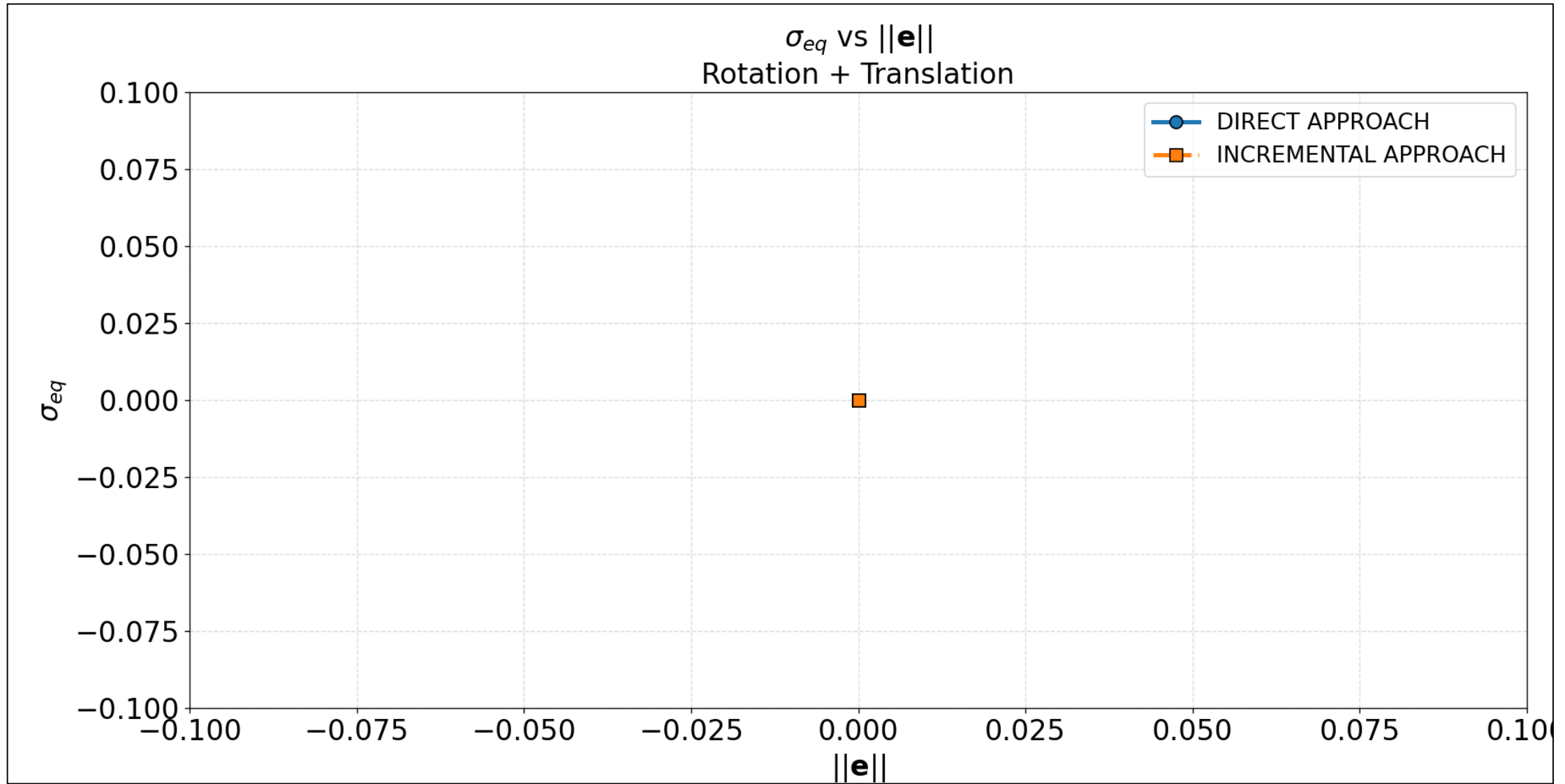
$$\det(F) = 1$$



Seq vs ||E|| plot



σ_{eq} vs $\|\mathbf{e}\|$ plot



MAP-3 PURE SHEAR

```
Time t = 0.99
=====
F =
[[1.  0.99 0. ]
 [0.  1.  0. ]
 [0.  0.  1. ]]

C =
[[1.  0.99 0. ]
 [0.99 1.9801 0. ]
 [0.  0.  1. ]]

E =
[[0.  0.495 0. ]
 [0.495 0.49005 0. ]
 [0.  0.  0. ]]

b =
[[1.9801 0.99 0. ]
 [0.99 1.  0. ]
 [0.  0.  1. ]]

e =
[[-1.11022302e-16  4.95000000e-01  0.00000000e+00]
 [ 4.95000000e-01 -4.90050000e-01  0.00000000e+00]
 [ 0.00000000e+00  0.00000000e+00  0.00000000e+00]]

S =
[[-85758.75  86625.  0. ]
 [ 86625.  0.  0. ]
 [ 0.  0.  0. ]]

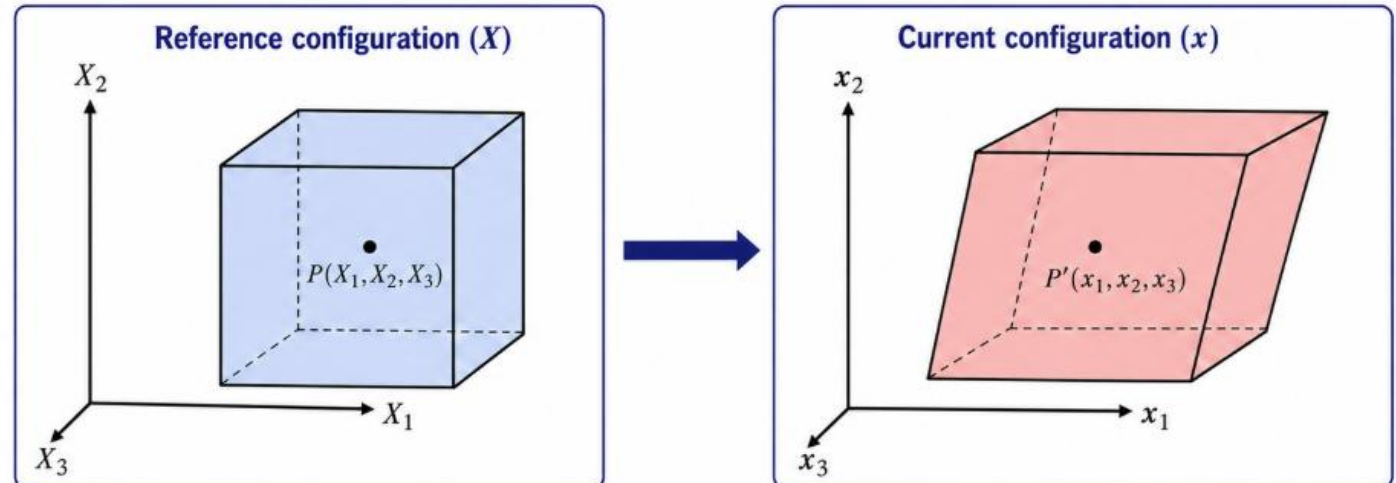
sigma =
[[85758.75  86625.  0. ]
 [86625.  0.  0. ]
 [ 0.  0.  0. ]]

Seq = 172818.50328180284
sigma_eq = 172818.50328180284
||E|| = 0.649618746362049
||e|| = 0.6496187463620491
```

CODE OUTPUT

Pure Shear Mapping

Simple shear deformation: points move in the x_1 -direction by an amount proportional to X_2 .
Shape changes, but the volume is preserved.



Mapping (simple shear):

$$\mathbf{x} = \chi(\mathbf{X}, t)$$

Component relations:

$$\begin{cases} x_1 = X_1 + t X_2 \\ x_2 = X_2 \\ x_3 = X_3 \end{cases}$$

Deformation gradient:

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \begin{bmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Deformation result

Simple shear: not a rigid-body motion

$$\det(\mathbf{F}) = 1$$

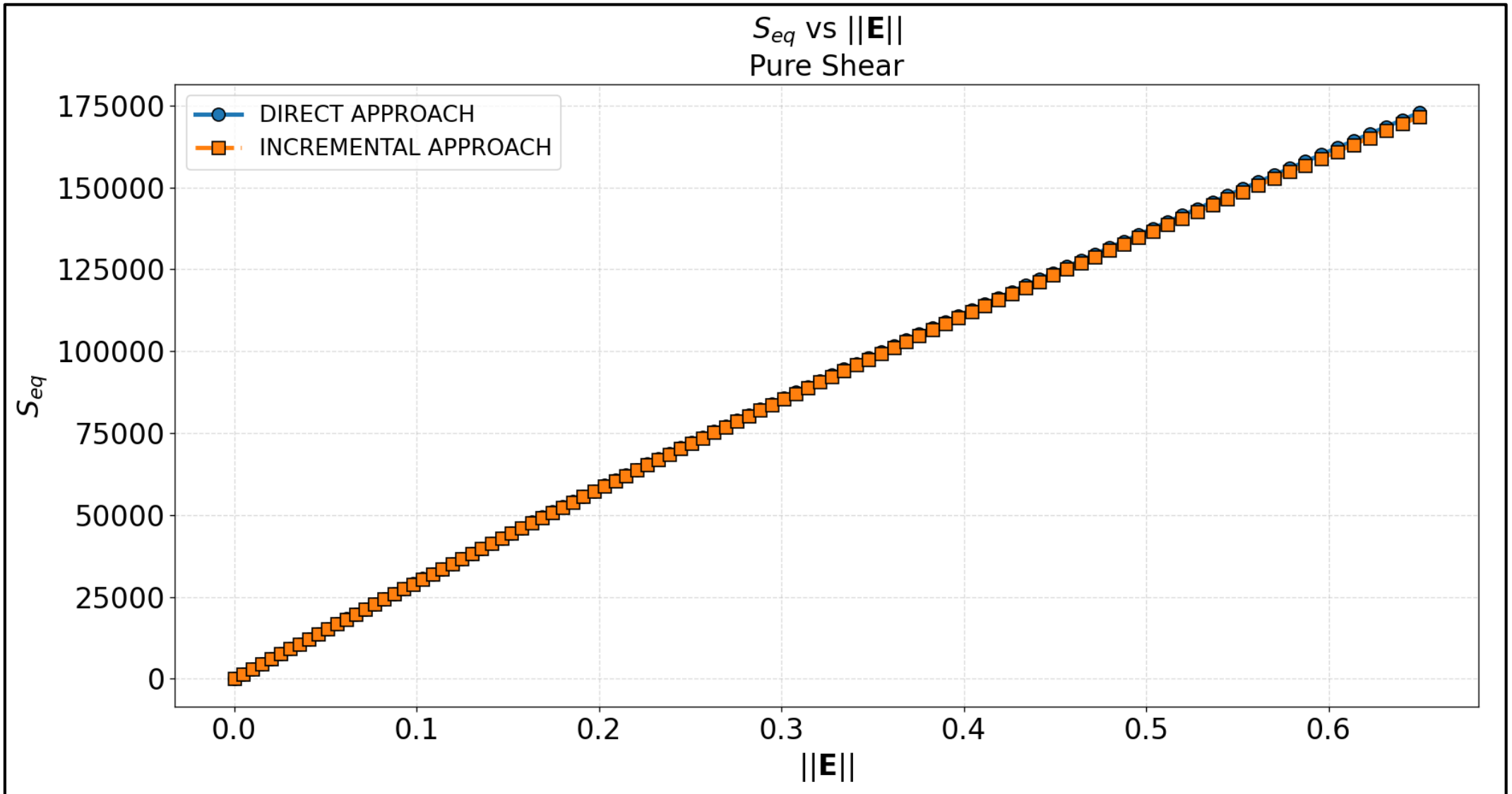
$$\mathbf{C} = \mathbf{F}^T \mathbf{F} \neq \mathbf{I}$$

$$\mathbf{E} = \frac{1}{2} (\mathbf{C} - \mathbf{I}) \neq \mathbf{0}$$

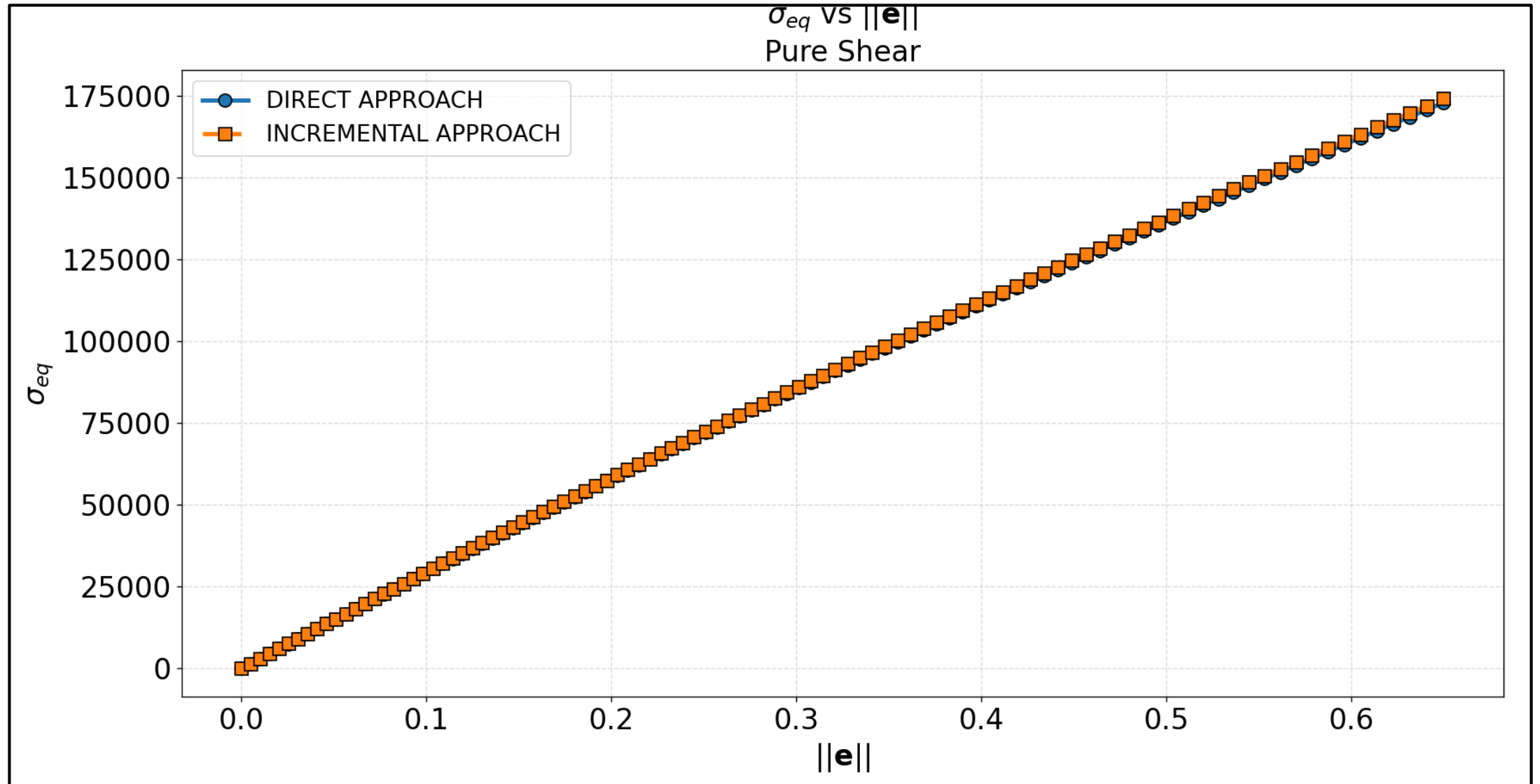
Right Cauchy-Green tensor:

$$\mathbf{C} = \begin{bmatrix} 1 & t & 0 \\ t & 1 + t^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Seq vs ||E|| plot



σ_{eq} vs $||\mathbf{e}||$ plot





MAP-4 GENERIC MAPPING

```

Time t = 0.99
=====
F =
[[2.69123447 0.      1.69123447]
 [0.      1.      2.31965778]
 [0.      0.      1.      ]]

C =
[[7.24274299 0.      4.55150851]
 [0.      1.      2.31965778]
 [4.55150851 2.31965778 9.24108626]]

E =
[[3.12137149 0.      2.27575426]
 [0.      0.      1.15982889]
 [2.27575426 1.15982889 4.12054313]]

b =
[[10.10301703 3.9230852 1.69123447]
 [ 3.9230852 6.38081222 2.31965778]
 [ 1.69123447 2.31965778 1.      ]]

e =
[[ 4.30965381e-01 4.39560983e-17 1.16753727e-01]
 [-1.14280538e-17 -1.11022302e-16 1.15982889e+00]
 [ 1.16753727e-01 1.15982889e+00 -2.88786404e+00]]

S =
[[ 71643.69349581 -43367.37880448 18695.59344209]
 [-43367.37880448 -102329.16361854 69009.81899448]
 [ 18695.59344209 69009.81899448 57750.      ]]

sigma =
[[317424.58656764 127551.11413083 54987.03953557]
 [127551.11413083 196404.68896227 75418.9417353 ]
 [ 54987.03953557 75418.9417353 21458.55390652]]

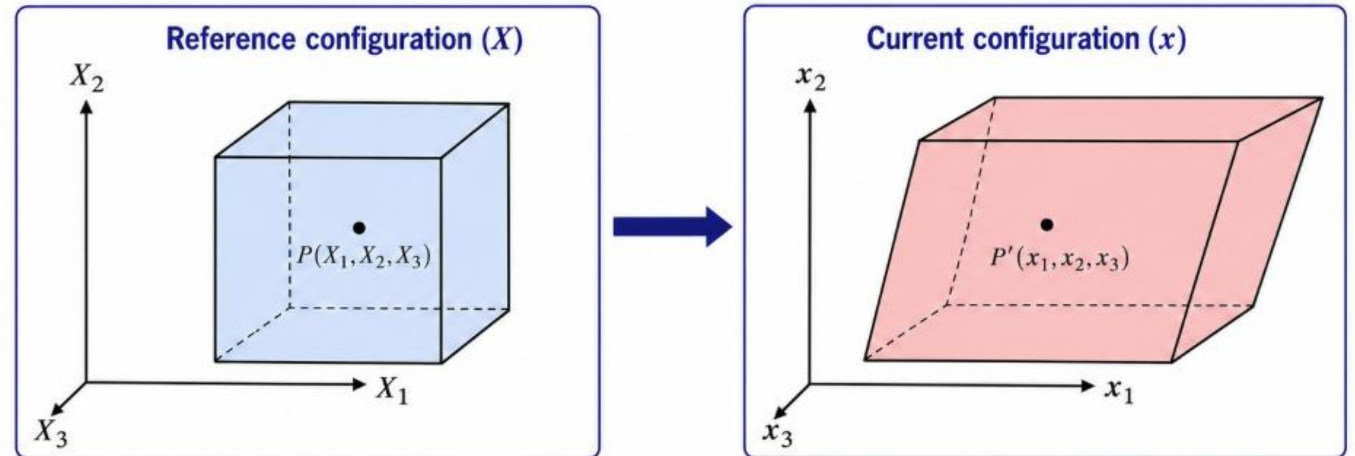
Seq = 221405.270503379
sigma_eq = 375987.4507937577
||E|| = 4.05230391989277
||e|| = 2.8636019259724605
=====
Map: Generic Map
Direct Approach vs Incremental Approach
Evaluation point: zeta = (1,1,1)

```

CODE OUTPUT

Generic Deformation Mapping

A general 3D deformation map combining exponential stretch in the X_1 -direction with X_3 -dependent coupling terms. The body deforms, so this is not a rigid-body motion.



Mapping

$$\mathbf{x} = \chi(\mathbf{X}, t)$$

Component relations:

$$\begin{cases} x_1 = e^t X_1 + (e^t - 1) X_3 \\ x_2 = X_2 + (e^t - e^{-t}) X_3 \\ x_3 = X_3 \end{cases}$$

Deformation gradient

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \begin{bmatrix} e^t & 0 & e^t - 1 \\ 0 & 1 & e^t - e^{-t} \\ 0 & 0 & 1 \end{bmatrix}$$

Deformation result

Not a rigid-body motion

$$\det(\mathbf{F}) = e^t$$

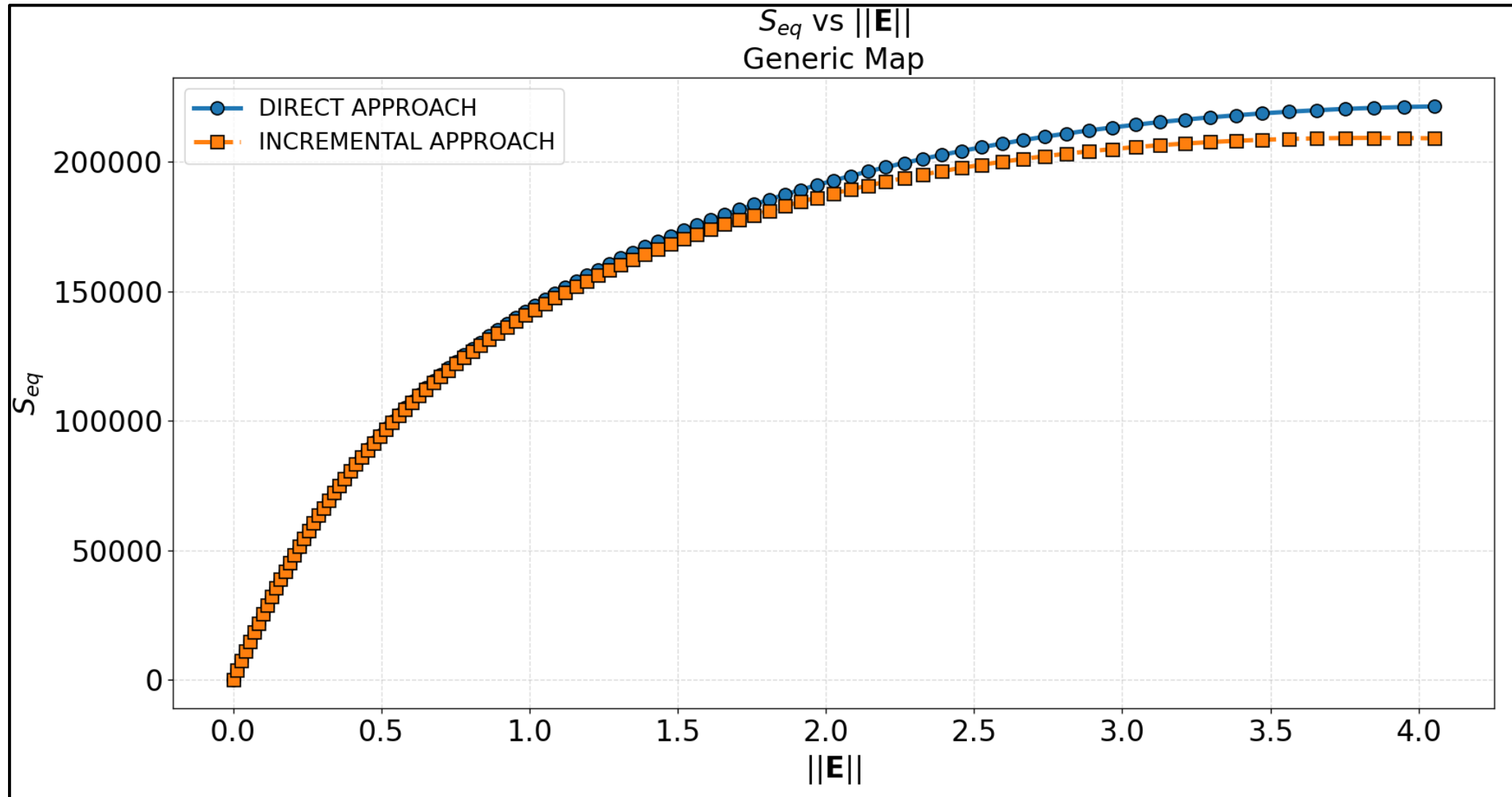
$$\mathbf{C} = \mathbf{F}^T \mathbf{F} \neq \mathbf{I}$$

$$\mathbf{E} = \frac{1}{2} (\mathbf{C} - \mathbf{I}) \neq \mathbf{0}$$

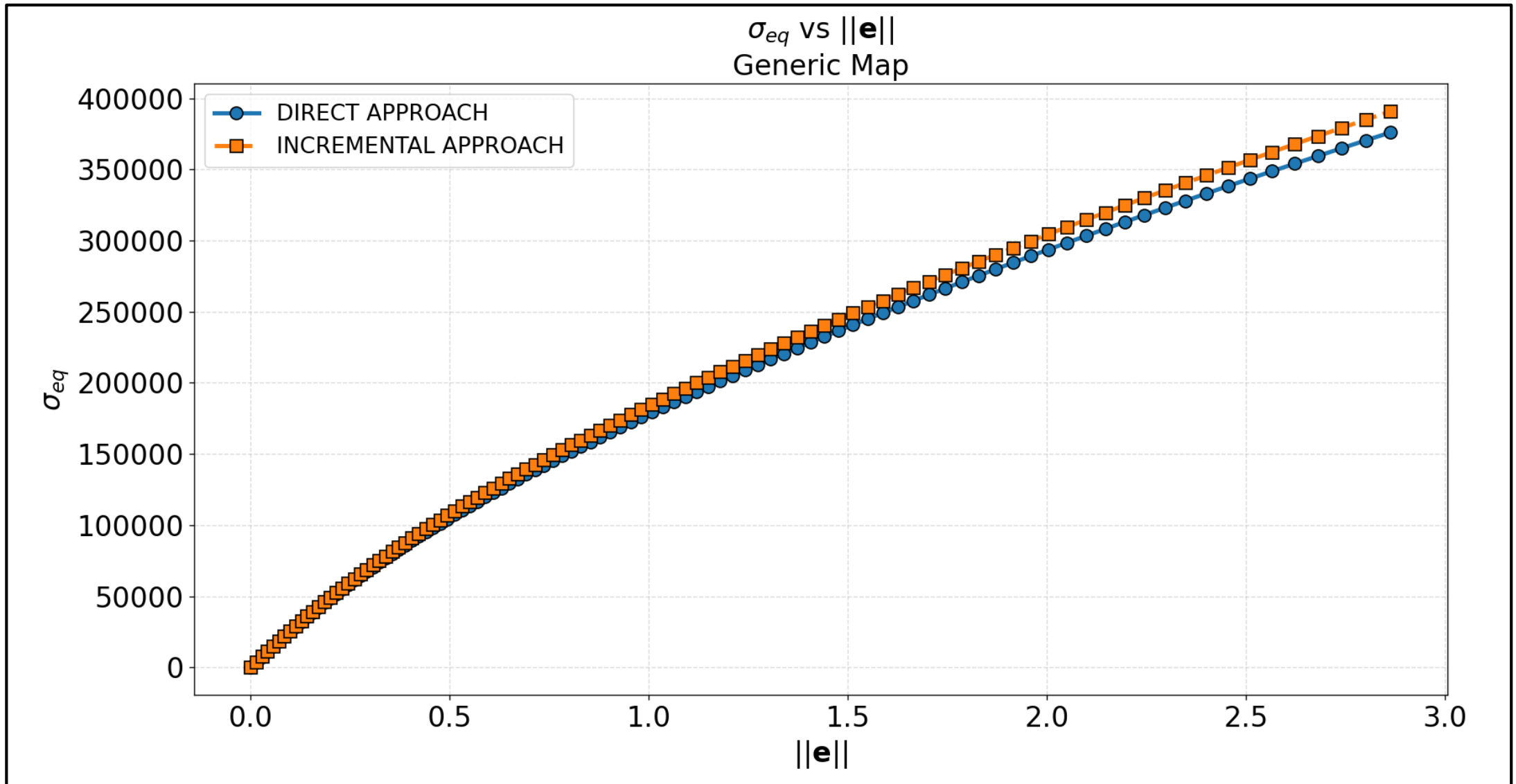
Volume changes and the deformation depends on X_3 .



Seq vs IIEI plot



σ_{eq} vs $\|e\|$ plot





MAP-5 INVERTED MAPPING

```

Time t = 0.99
=====
F =
[[2.69123447 0.      1.69123447]
 [0.      1.      2.31965778]
 [0.      0.      1.      ]]

C =
[[7.24274299 0.      4.55150851]
 [0.      1.      2.31965778]
 [4.55150851 2.31965778 9.24108626]]

E =
[[3.12137149 0.      2.27575426]
 [0.      0.      1.15982889]
 [2.27575426 1.15982889 4.12054313]]

b =
[[10.10301703 3.9230852 1.69123447]
 [ 3.9230852 6.38081222 2.31965778]
 [ 1.69123447 2.31965778 1.      ]]

e =
[[ 4.30965381e-01  4.39560983e-17  1.16753727e-01]
 [-1.14280538e-17 -1.11022302e-16  1.15982889e+00]
 [ 1.16753727e-01  1.15982889e+00 -2.88786404e+00]]

S =
[[ 71643.69349581 -43367.37880448 18695.59344209]
 [-43367.37880448 -102329.16361854 69009.81899448]
 [ 18695.59344209 69009.81899448 57750.      ]]

sigma =
[[317424.58656764 127551.11413083 54987.03953557]
 [127551.11413083 196404.68896227 75418.9417353 ]
 [ 54987.03953557 75418.9417353 21458.55390652]]

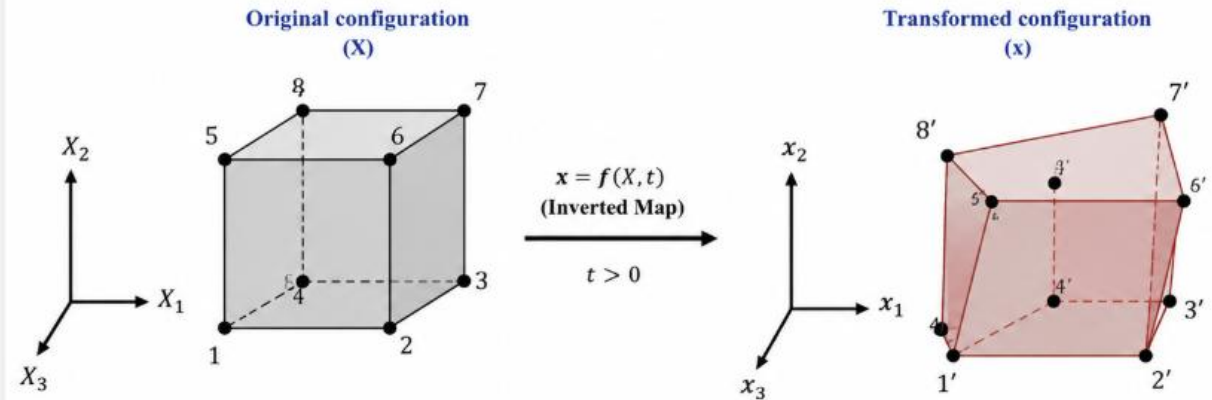
Seq = 221405.270503379
sigma_eq = 375987.4507937577
||E|| = 4.05230391989277
||e|| = 2.8636019259724605
=====
Map: Inverted Map
Direct Approach vs Incremental Approach
Evaluation point: zeta = (1,1,1)
=====

```

CODE OUTPUT

Map 5: Inverted (Orientation-Reversing) Map

This mapping includes a reflection (inversion) combined with an exponential scaling.
It reverses the orientation of the element ($\det F < 0$ for $t > 0$).



Inverted Map (transformation):

$$\mathbf{x} = \mathbf{f}(\mathbf{X}, t)$$

$$\begin{aligned} x_1 &= e^t X_1 + (e^t - 1) X_3 \\ x_2 &= X_2 + (e^t - e^{-t}) X_3 \\ x_3 &= X_3 \end{aligned}$$

In component form:

$$\left. \begin{aligned} x_1 &= e^t X_1 + (e^t - 1) X_3 \\ x_2 &= X_2 + (e^t - e^{-t}) X_3 \\ x_3 &= X_3 \end{aligned} \right\} \text{ or } x_i = f_i(X_1, X_2, X_3, t) \text{ for } i = 1, 2, 3$$

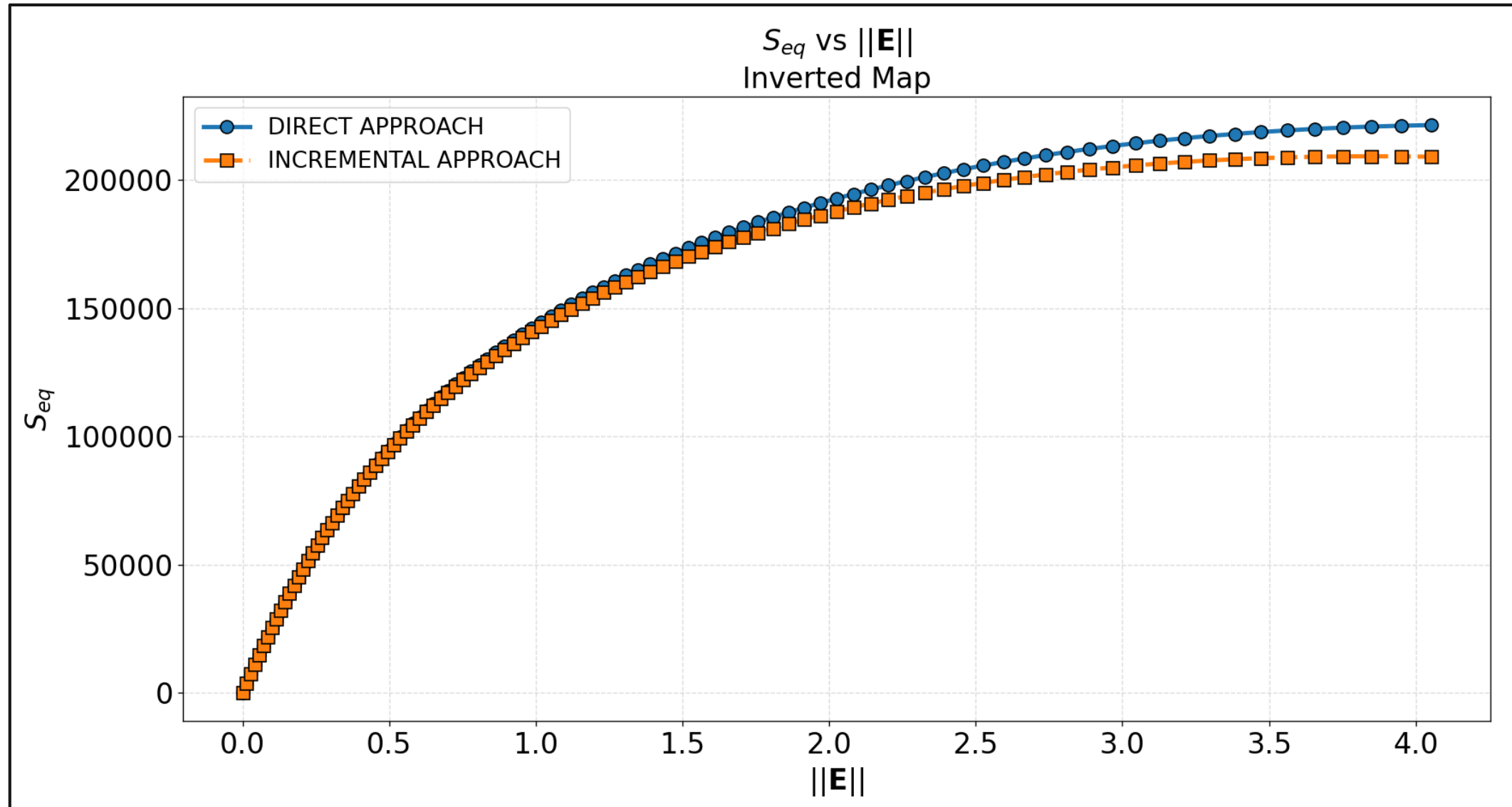
Matrix form:

$$\mathbf{x} = \mathbf{A}(t) \mathbf{X}, \quad \text{where } \mathbf{A}(t) = \begin{bmatrix} e^t & 0 & (e^t - 1) \\ 0 & 1 & (e^t - e^{-t}) \\ 0 & 0 & 1 \end{bmatrix}$$

Notes:

- Orientation is reversed ($\det \mathbf{A}(t) = -e^{2t} < 0$ for $t > 0$).
- $e^t > 1$ for $t > 0$, so the element is stretched and inverted.

Seq vs IIEI plot



σ_{eq} vs $\|e\|$ plot

